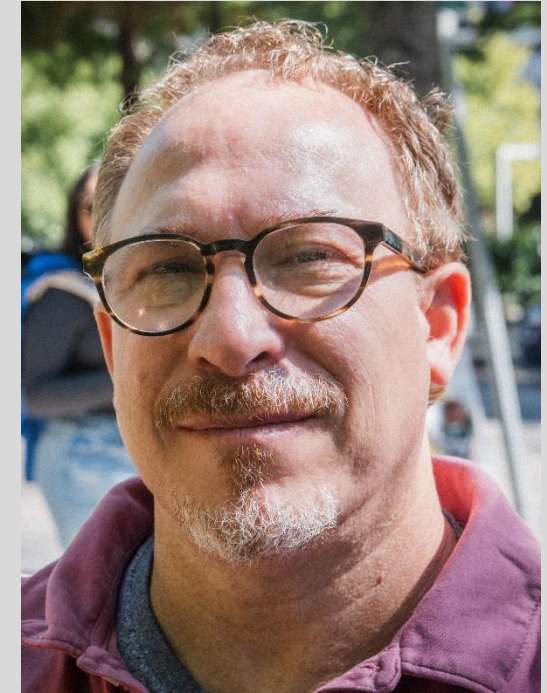


Cranial Nerve Nuclei: Cranial Nerves V and VII-XII

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Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

1. Students will be able to identify the cranial nerves for all functions (aside from olfaction, vision, and eye movements) and the particular information each nerve relays.
2. Students will be able to identify the relevant cranial nerve nuclei and their relation to the particular nerves.
3. Students will be able to identify the overall organization of the cranial nerve nuclei and each nuclei's general position.
4. Students will be able to locate injuries to these pathways given symptoms affecting specific capabilities.

Cranial Nerve Nuclei (Motor)

Nucleus	nerve	function	effectors
abducens	VI	somatic motor	eyes
oculomotor	III	somatic motor	eyes
trochlear	IV	somatic motor	eyes
hypoglossal	XII	somatic motor	tongue
ambiguus*	IX, XI	branchial motor	swallowing
	X	branchial motor	throat, vocals
facial motor	VII	branchial motor	face, upper neck
spinal accessory**	XI	branchial motor	neck, shoulders
trigeminal motor	V	branchial motor	jaw, upper neck
dorsal vagus	X	visceral efferent	gastrointestinal, heart
Edinger Westphal	III	visceral efferent	pupil constriction
inferior salivary	X	visceral efferent	glands (mouth)
superior salivary	VII	visceral efferent	glands (mouth, tears)

* ambiguus has a brachial component and a visceral efferent component to inhibit heart rate

**spinal accessory is sometimes classified as brachial and sometimes as somatic motor.

Cranial Nerve Nuclei (Sensory)

Nucleus	nerve	function	sensors
gustatory	VII	visceral sensory	taste
	IX, X	visceral sensory	taste
solitary	IX	visceral sensory	blood pressure, [O2]
	X	visceral sensory	respiratory, cardio, digestive
trig. mesencephalic	V	somatic sensory	proprioception (jaw) ** <i>exception – 1st order afferents</i>
trig. pontine	V	somatic sensory	touch/proprioception (face)
trig. spinal	V	somatic sensory	pain/temperature (face, oral cavity)
	VII	somatic sensory	pain/temperature (ear)
	IX	somatic sensory	pain/temperature (throat, vocals)
	X	somatic sensory	pain/temperature (ear, pharynx)
vestibular	VIII	special sensory	head motion / orientation
cochlear	VIII	special sensory	hearing

Brainstem nuclei position	midbrain	Edinger Westphal	III	pupil constriction	visceral efferent
		oculomotor	III	eyes	somatic motor
		trig. mesencephalic	V	jaw	somatic sensory
		trochlear	IV	eyes	somatic motor
	pons	trigeminal motor	V	jaw, upper neck	branchial motor
		trigeminal pontine	V	face	somatic sensory
		abducens	VI	eyes	somatic motor
		facial motor	VII	face, upper neck	branchial motor
		superior salivary	VII	mouth	visceral efferent
		vestibular	VIII	head motion / orientation	special sensory
		cochlear	VIII	hearing	special sensory
		inferior salivary	IX	parotid gland	visceral efferent
		solitary gustatory	{ VII	taste	visceral sensory
			{ IX, X	taste	visceral sensory
	medulla	solitary	{ IX	blood pressure, [O ₂]	sensory
			{ X	respiratory, cardio, digestive	visceral sensory
		hypoglossal	XII	tongue	somatic motor
		ambiguus*	{ IX, XI	swallowing	branchial motor
			{ X	throat, vocals	branchial motor
		spinal accessory**	XI	neck, shoulders	branchial motor
		dorsal vagus	X	gastrointestinal, heart	visceral efferent
		trigeminal spinal	{ V	face, oral cavity	somatic sensory
			{ VII	ear	somatic sensory
			{ IX	throat, vocals	somatic sensory
			{ X	ear, pharynx	somatic sensory

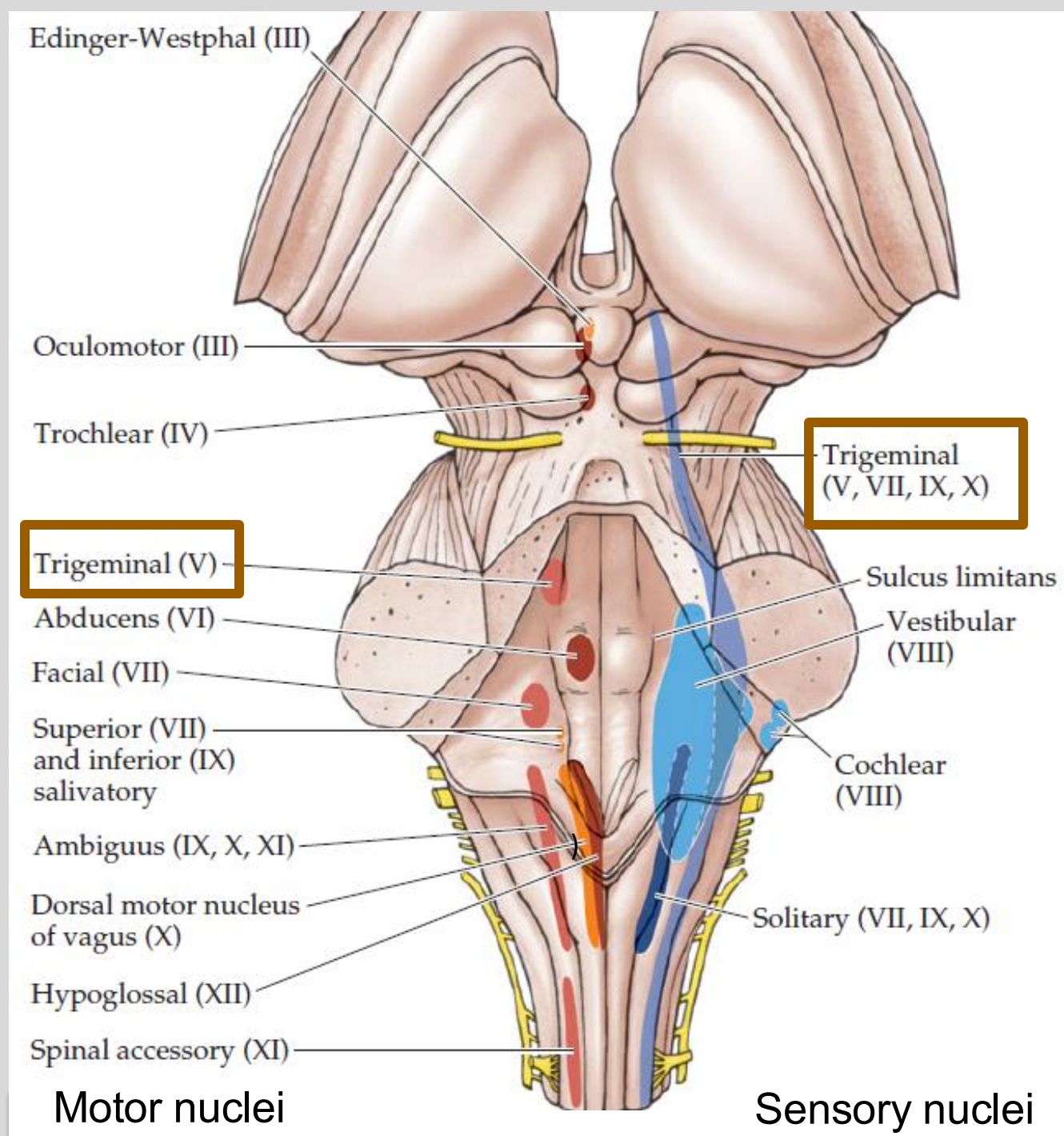
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<i>cranial nerve nuclei summary</i>	<u>SOMATIC MOTOR</u>	<u>BRANCHIAL MOTOR</u>	<u>VISCERAL MOTOR</u>	<u>VISCERAL SENSORY</u>	<u>SOMATIC SENSORY</u>	<u>SPECIAL SENSORY</u>
V		<i>trigeminal motor</i> → jaw	← Easy enough →		jaw proprio → <i>trig. mesen.</i> face touch & proprio → <i>trig. pontine</i> face pain & temp → <i>trig spinal</i>	
VII		<i>facial motor</i> → face & stapedius	<i>sup. salivatory</i> → tears & saliva	ant. tongue taste → <i>gustatory</i>	ear → <i>trig. pontine</i> <i>trig. spinal</i>	
VIII	1. The CNs are always sensory OR motor 2. The CNs can be sensory AND motor					orientation → <i>vestibular</i> hearing → <i>cochlear</i>
IX		<i>ambiguus</i> → stylopharyngeus (swallow)	<i>inf. salivatory</i> → parotid gland (saliva)	posterior tongue taste → <i>gustatory</i> carotid baroreceptors → <i>solitary</i>	posterior tongue, pharynx and ear → <i>trig. pontine</i> <i>trig. spinal</i>	
X		<i>ambiguus</i> → pharynx&larynx (vocalize, swallow, gag reflex)	<i>dorsal motor vagus</i> → “rest and digest” autonomics	viscerosensory & aortic baroreceptors → <i>solitary</i>	ear → <i>trig. pontine</i> <i>trig. spinal</i>	
XI		<i>spinal accessory</i> → scm and upper trapezius				
XII	<i>hypoglossal</i> → tongue					

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VIII						orientation → <i>vestibular</i> hearing → <i>cochlear</i>
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XI		<i>spinal accessory</i> → scm and upper trapezius				
XII	<i>hypoglossal</i> → tongue					

CN V (trigeminal)



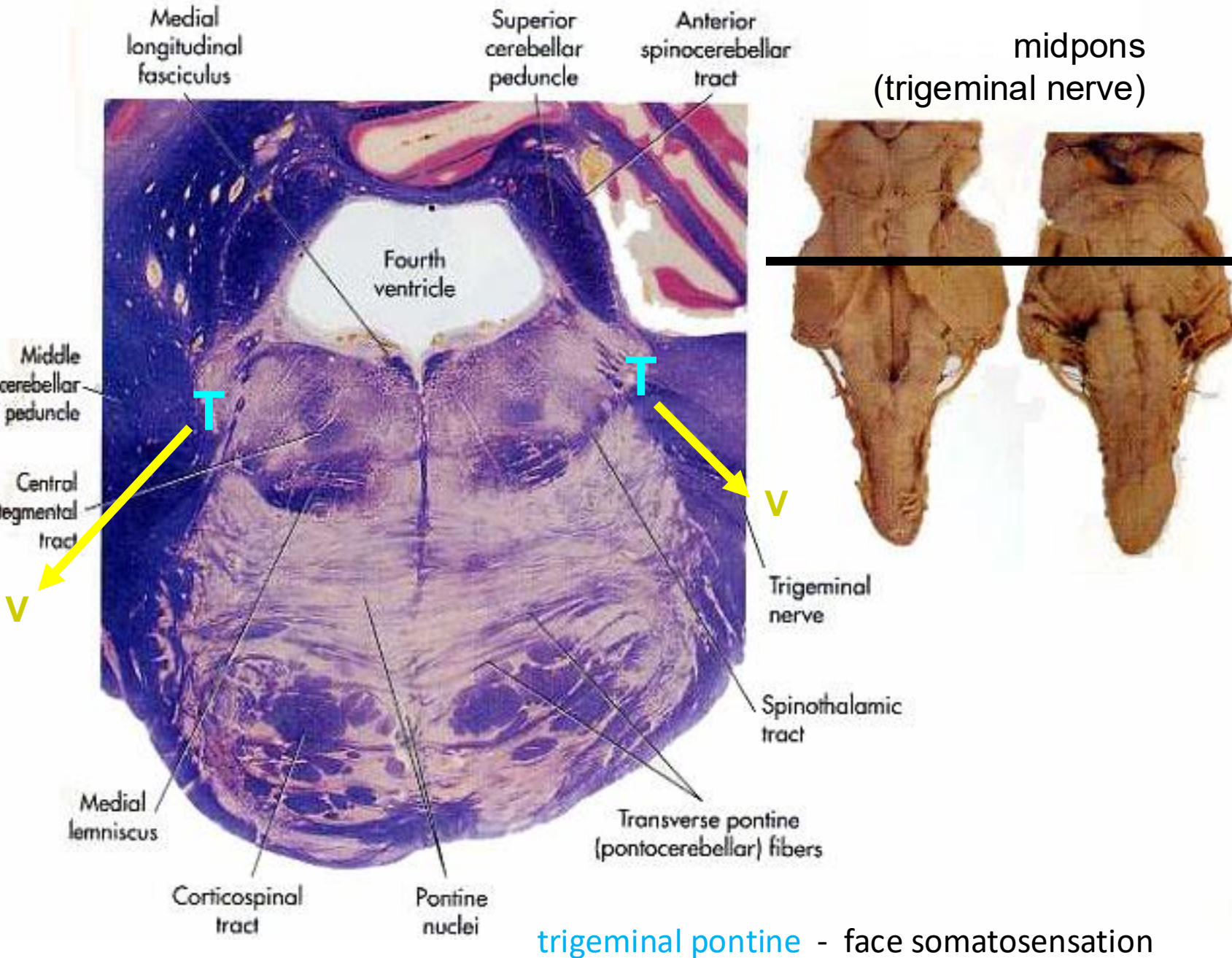
CN V (trigeminal)

Three major divisions: V1 **ophthalmic** – somatosensation
V2 **maxillary** – somatosensation
V3 **mandibular** – somatosensation AND motor

trigeminal pontine nuclei ← V1-3 touch & proprioception
trigeminal spinal nuclei ← V1-3 pain & temperature } ipsilateral face incl. eye
& anterior 2/3 tongue

trigeminal pontine and spinal nuclei also receive skin feedback
from CN VII (ear), CN IX (posterior 1/3 tongue, ear, throat), and CN X (ear)

trigeminal mesencephalic nuclei ← V3 jaw proprioception (the nuclei is 1st order neuron)
trigeminal motor nuclei → V3 muscles of mastication

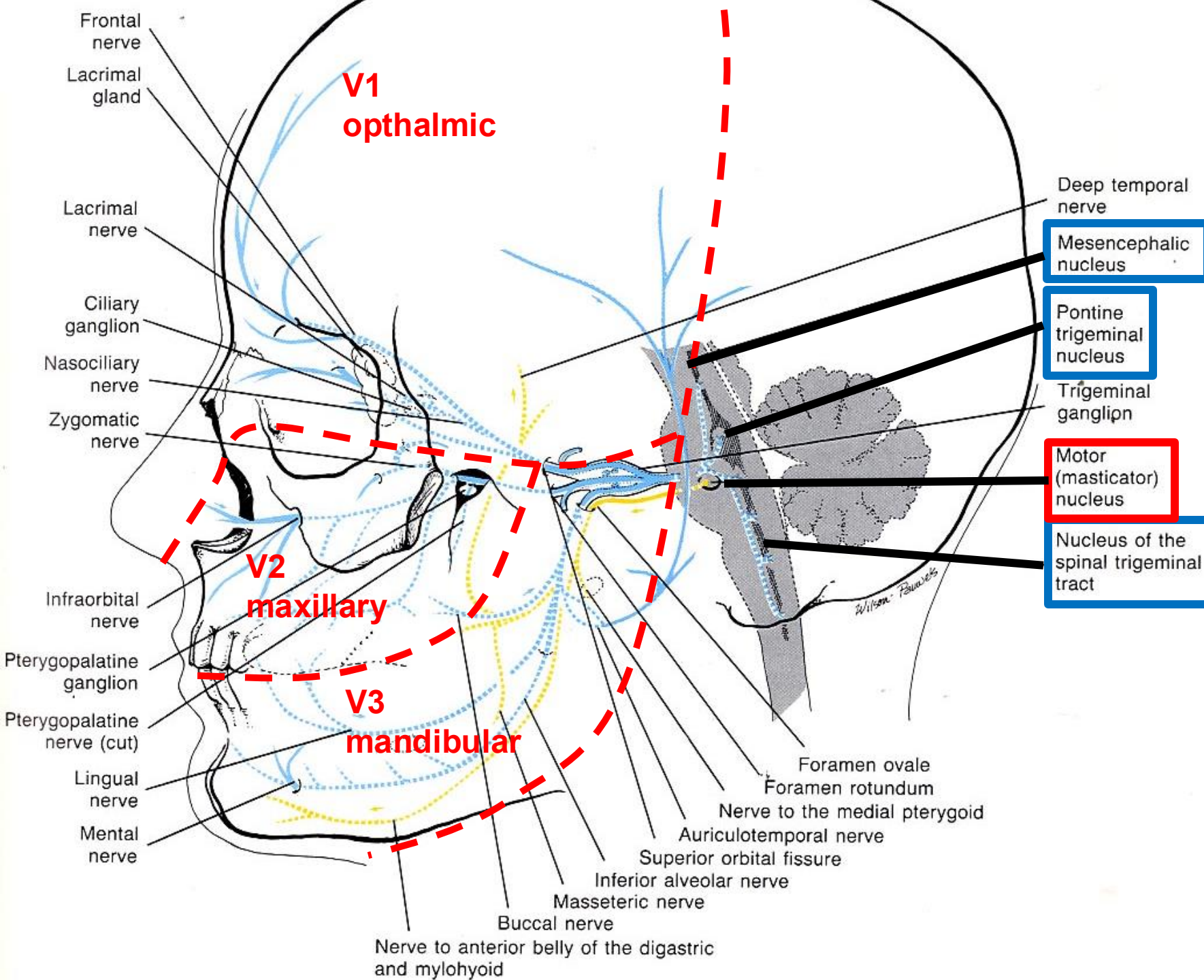


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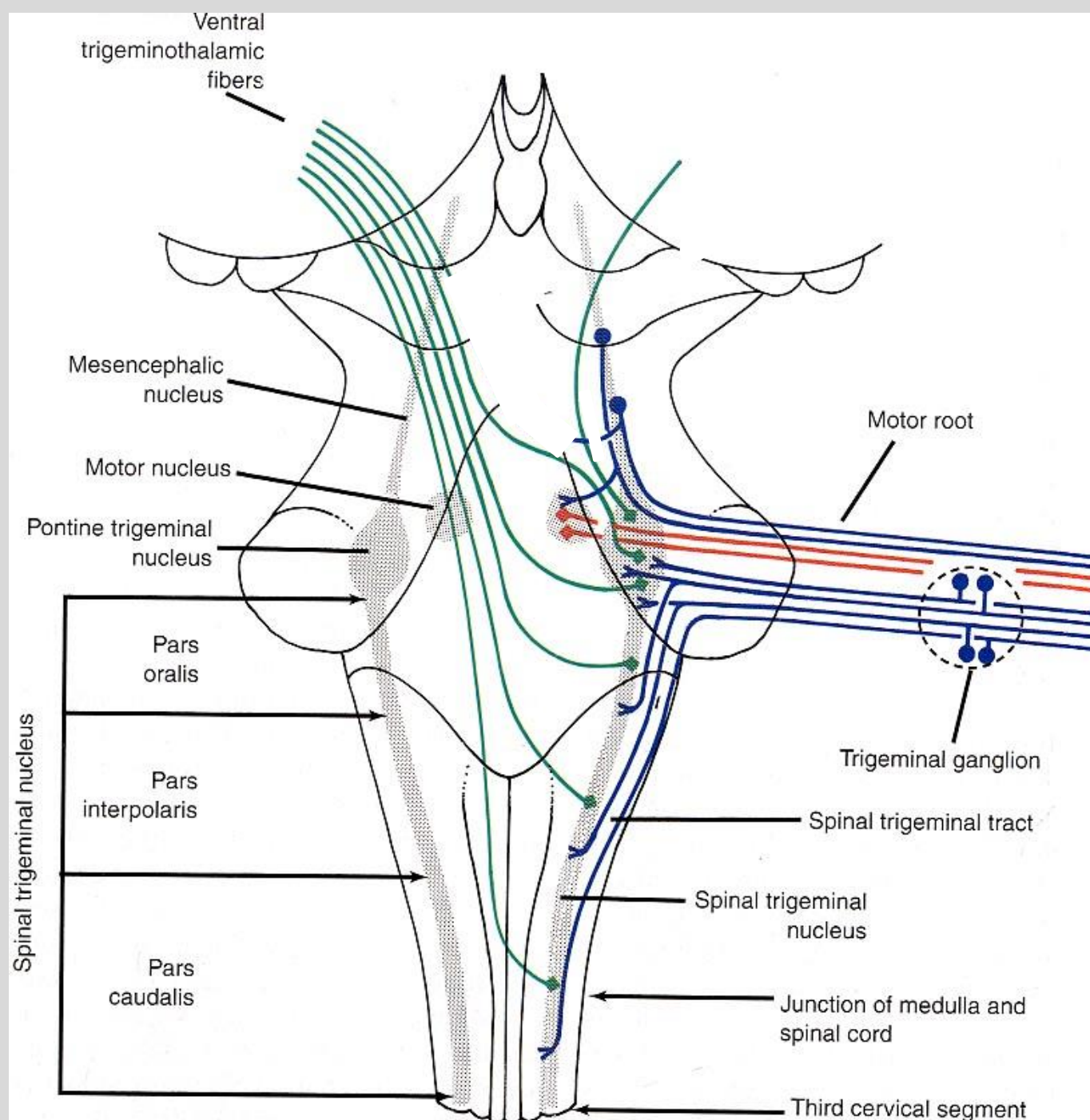
CN V (trigeminal)

CN V (trigeminal)



*Note: skin sense of the back
of head is handled by spinal
nerves*

CN V (trigeminal)



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Principle and spinal trigeminal nuclei project to the **contralateral** thalamus

Pain/temp fibers descend to spinal trigeminal nucleus

Lesion at upper spinal level would blunt pain/temp for ipsilateral face but not touch or proprioception!

CN V Clinical correlations

Fracture of facial bones → Selective anesthesia

Trigeminal Neuralgia (nerve pain)

Superior Cerebellar Artery compressing sensory root

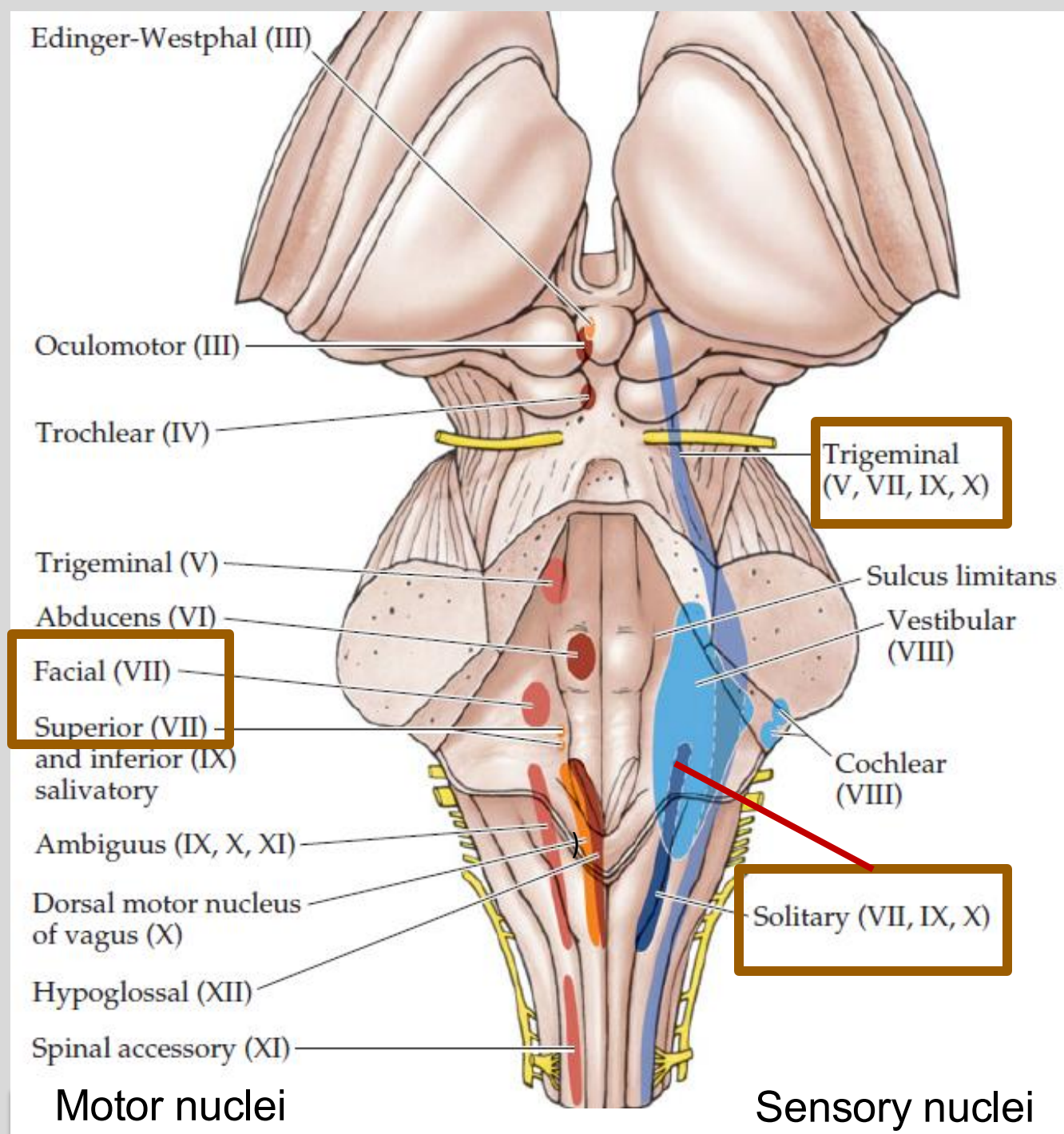
→ Demyelination damage

“Suicide Disease” –sudden pain to eyes, lips, and head
100s/per day

Loss of Corneal Reflex (blinking of eyelids)

Afferent limb of CN V-1

CN VII (facial)



CN VII (facial)

facial motor nucleus → facial expression muscles, diaphragm, stapedius (middle ear)

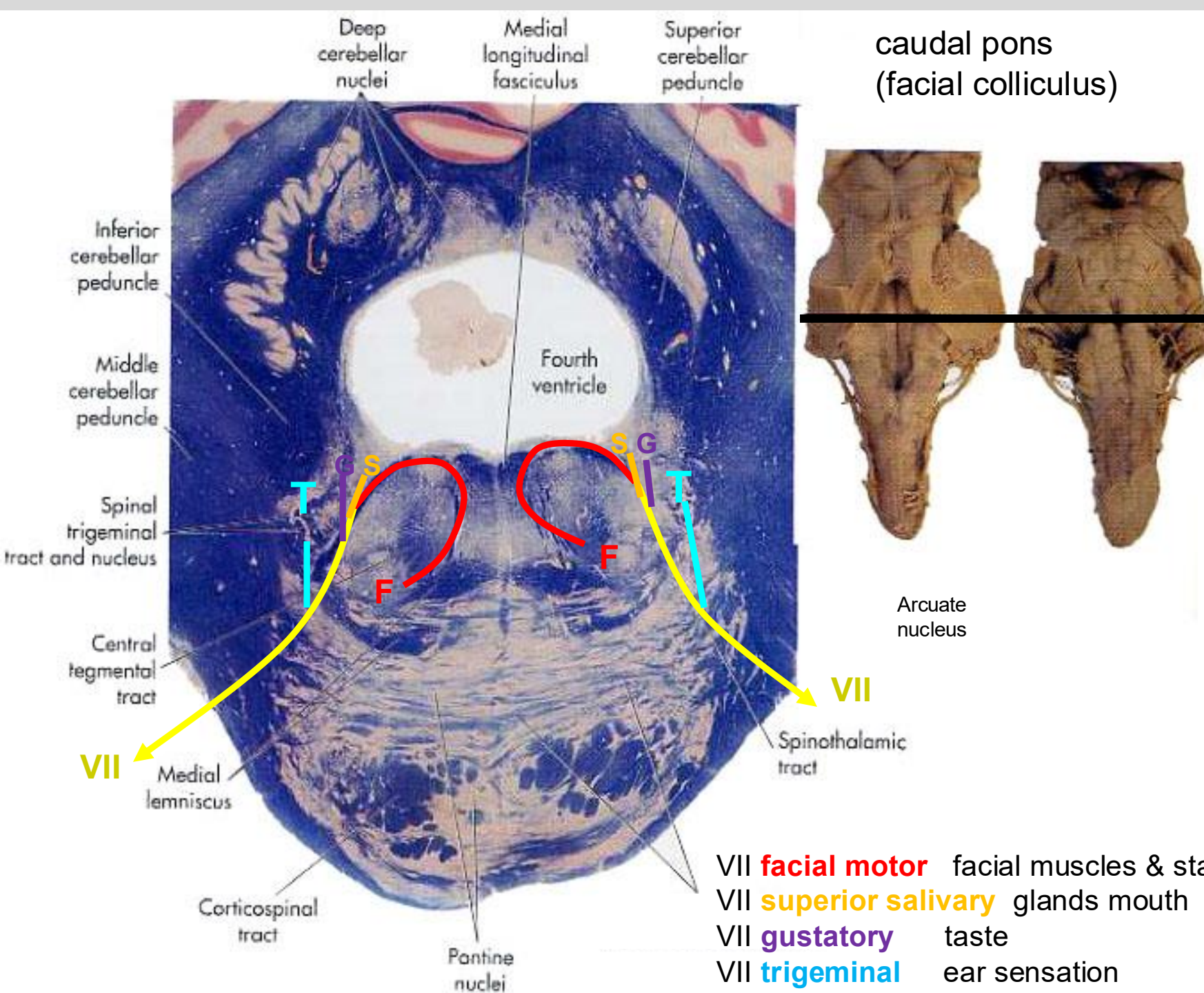
sup. salivatory nucleus → lacrimal (tears), submandibular and sublingual glands (saliva)

gustatory nucleus ← taste (anterior 2/3 of tongue)

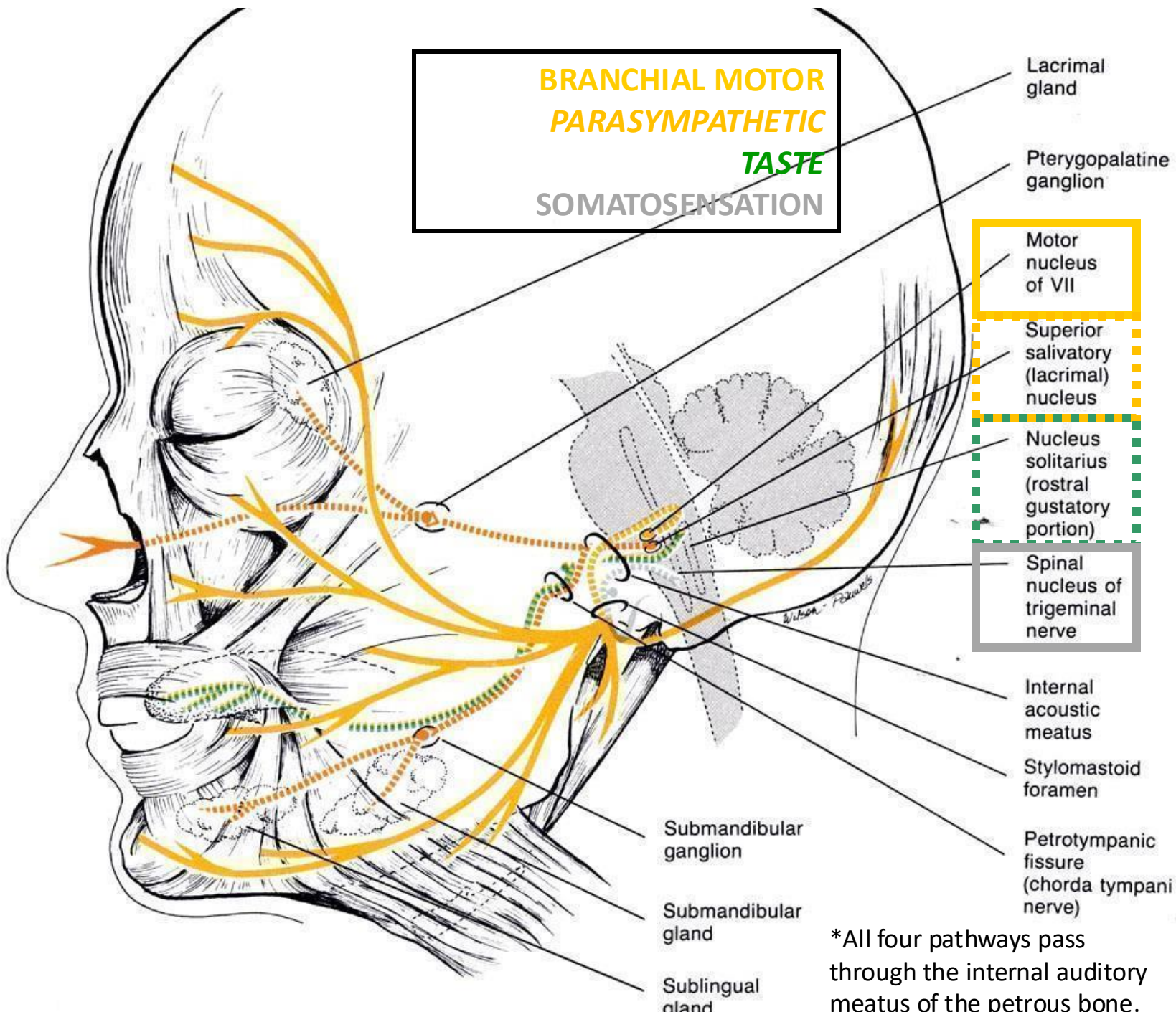
trigeminal nucleus ← posterior surface of external ear skin senses

“Enjoying a Greek olive, not chewing, just tasting and smiling and drooling a bit, and the tears well up its just so tasty!! and then my wife drags me out of the store by the ear.”

CN VII (facial)

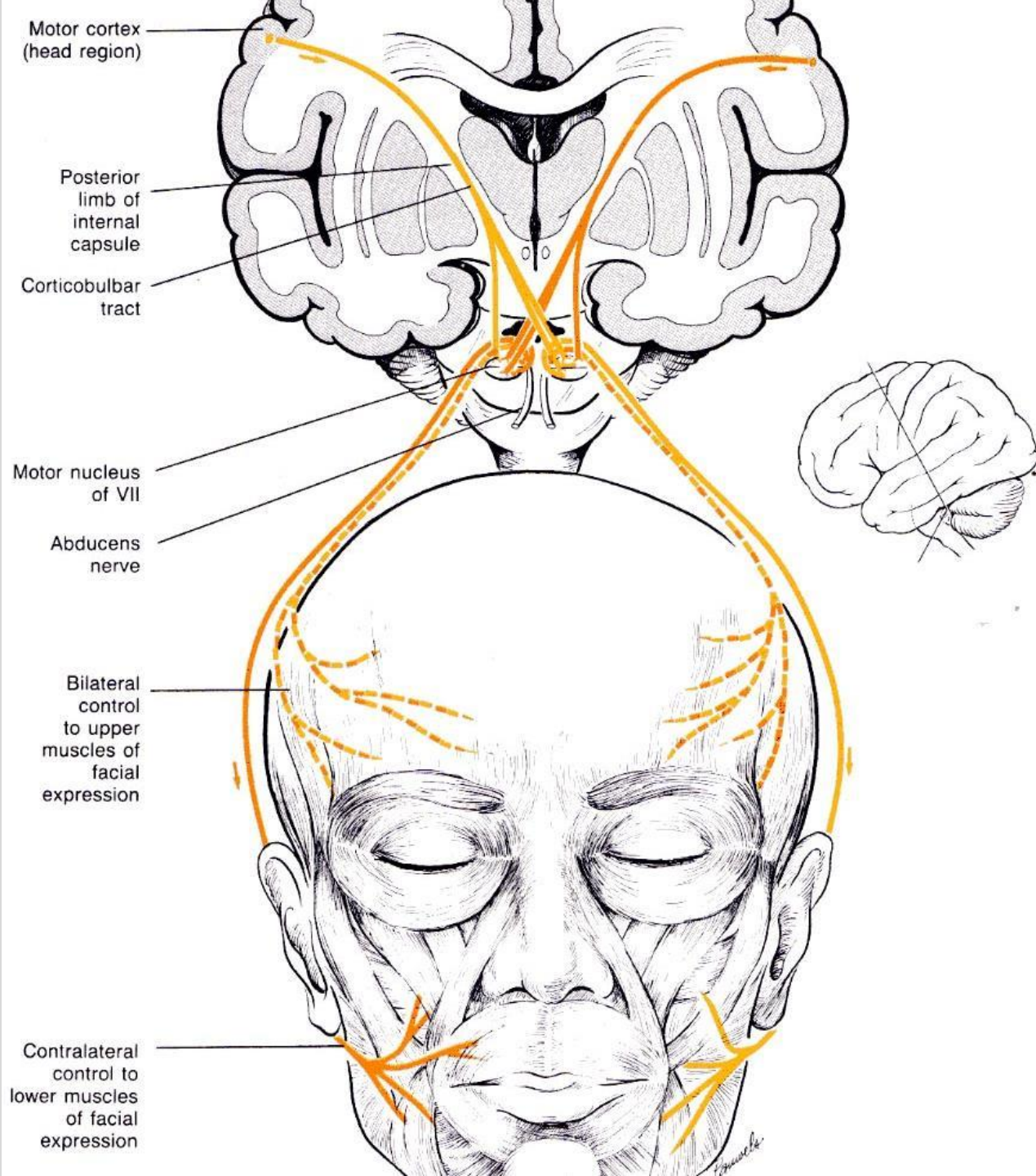


CN VII (facial)



*All four pathways pass through the internal auditory meatus of the petrous bone.

CN VII (facial)



forehead / eyebrow /

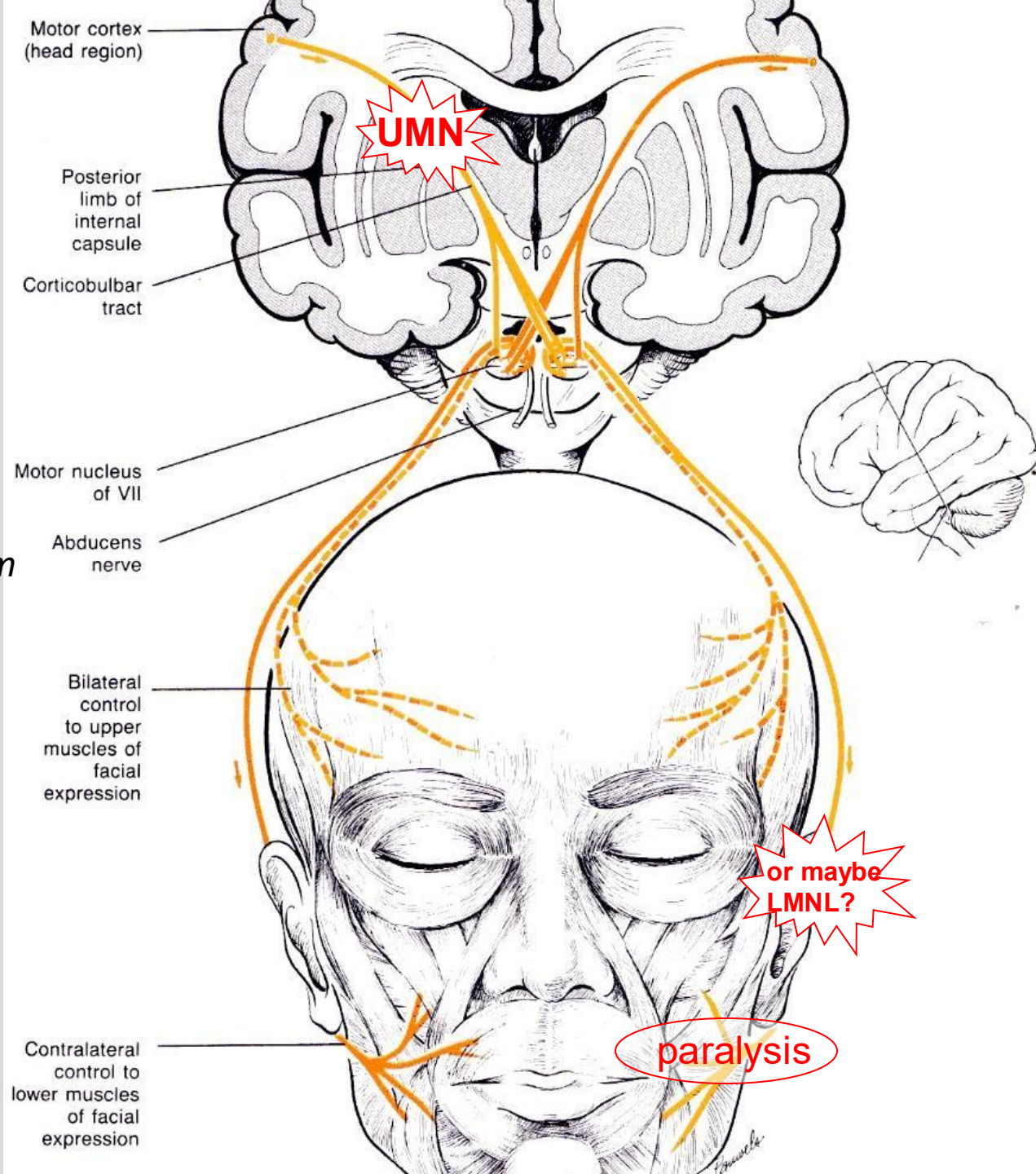
*cortical motor neurons project **BILATERALLY** to facial motor nuclei*

*Cheek cortical motor neurons project **contralaterally** to facial motor nuclei*

CN VII (facial)

Clinical correlation

*Facial paralysis in lower
part of face is either from
UMN lesion or CN VII
inferior branch lesion*



Clinical correlation

CN VII vulnerable in middle ear

Bell's Palsy

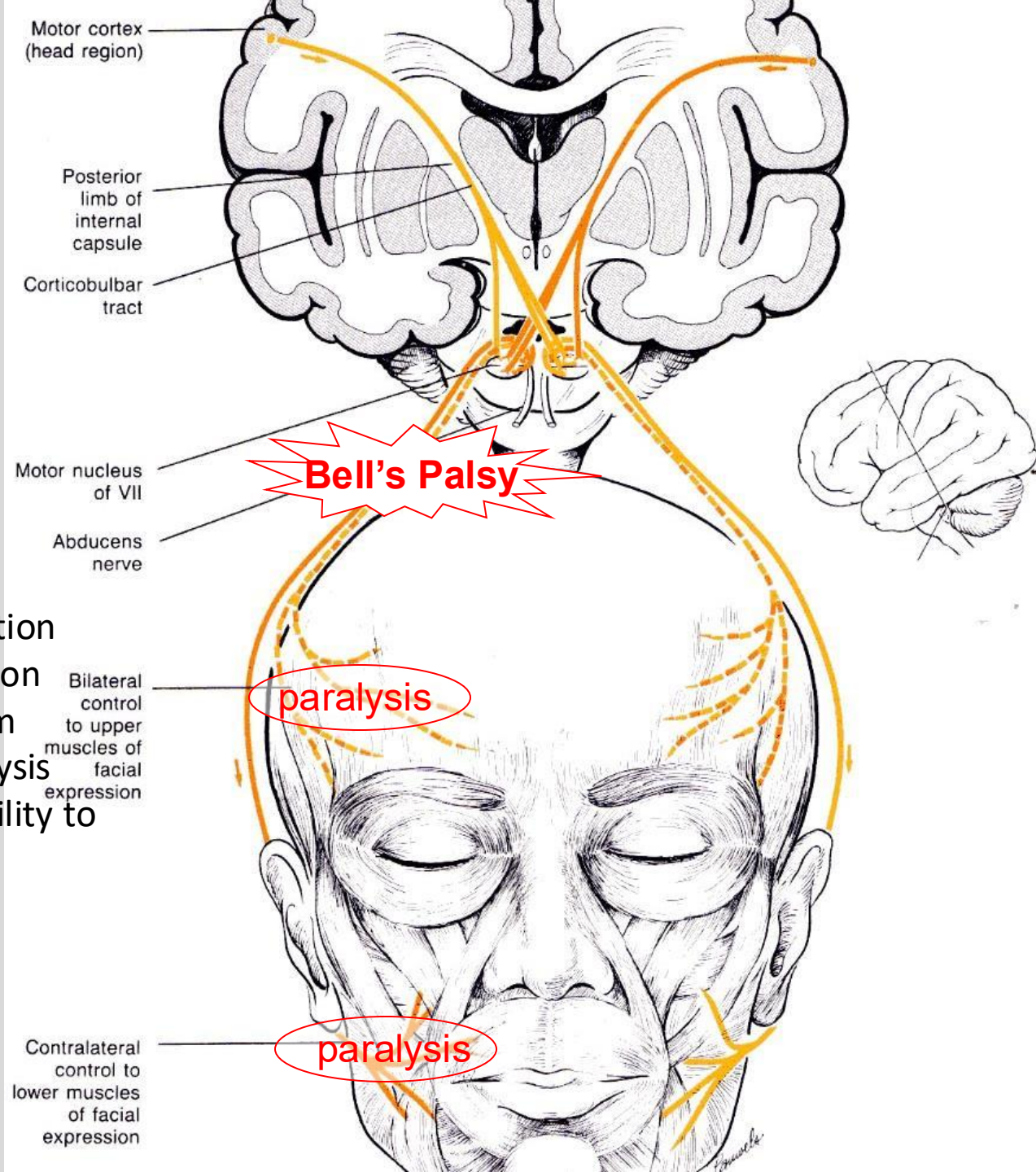
can also cause:

diminished taste sensation

and salivation

excessive loudness from stapedius muscle paralysis

eye dryness (from inability to close eye)

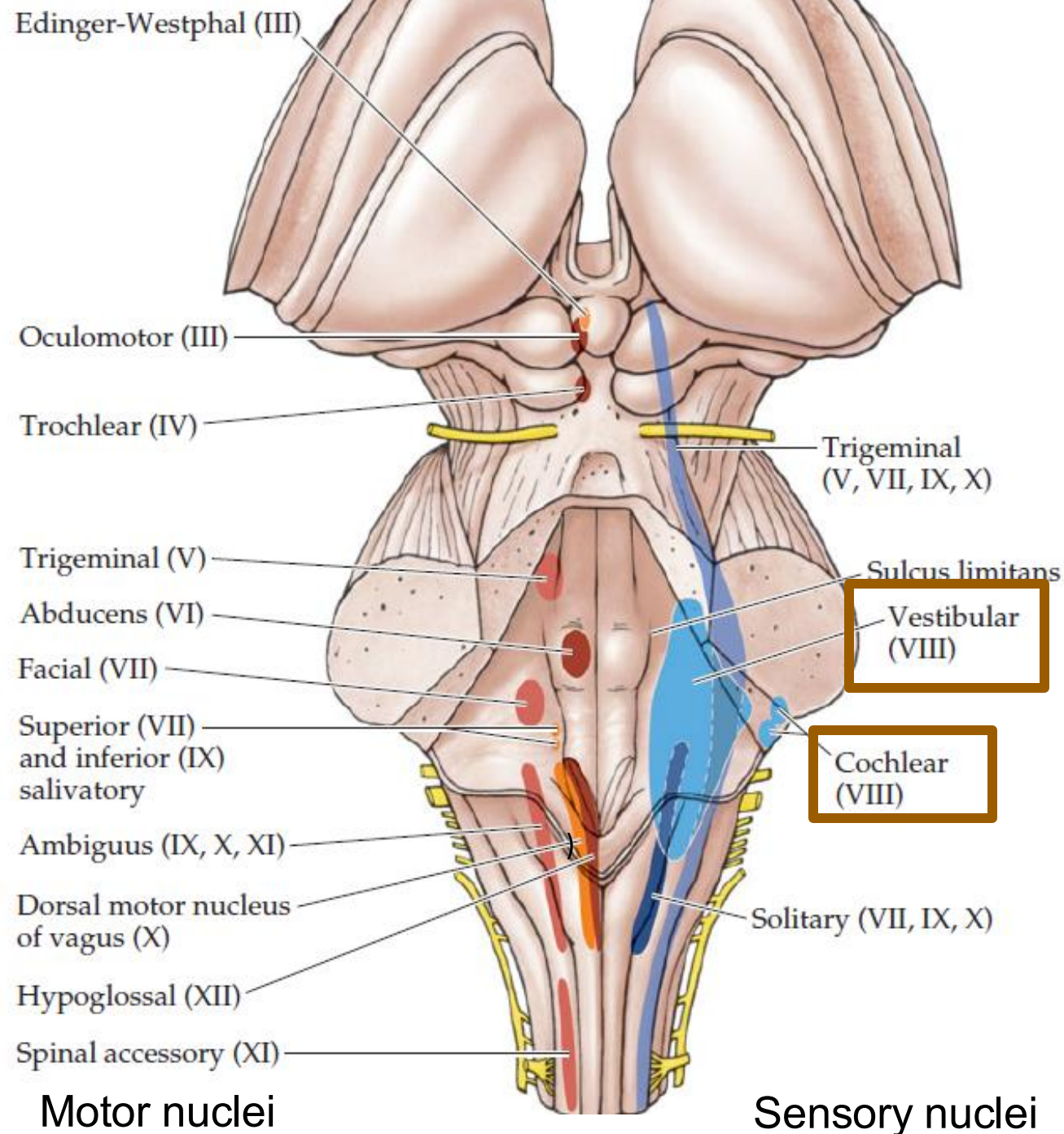


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CN VII (facial)

CN VIII (vestibulocochlear)



CN VIII (vestibulocochlear)

cochlear nucleus ← organ of Corti (hearing)

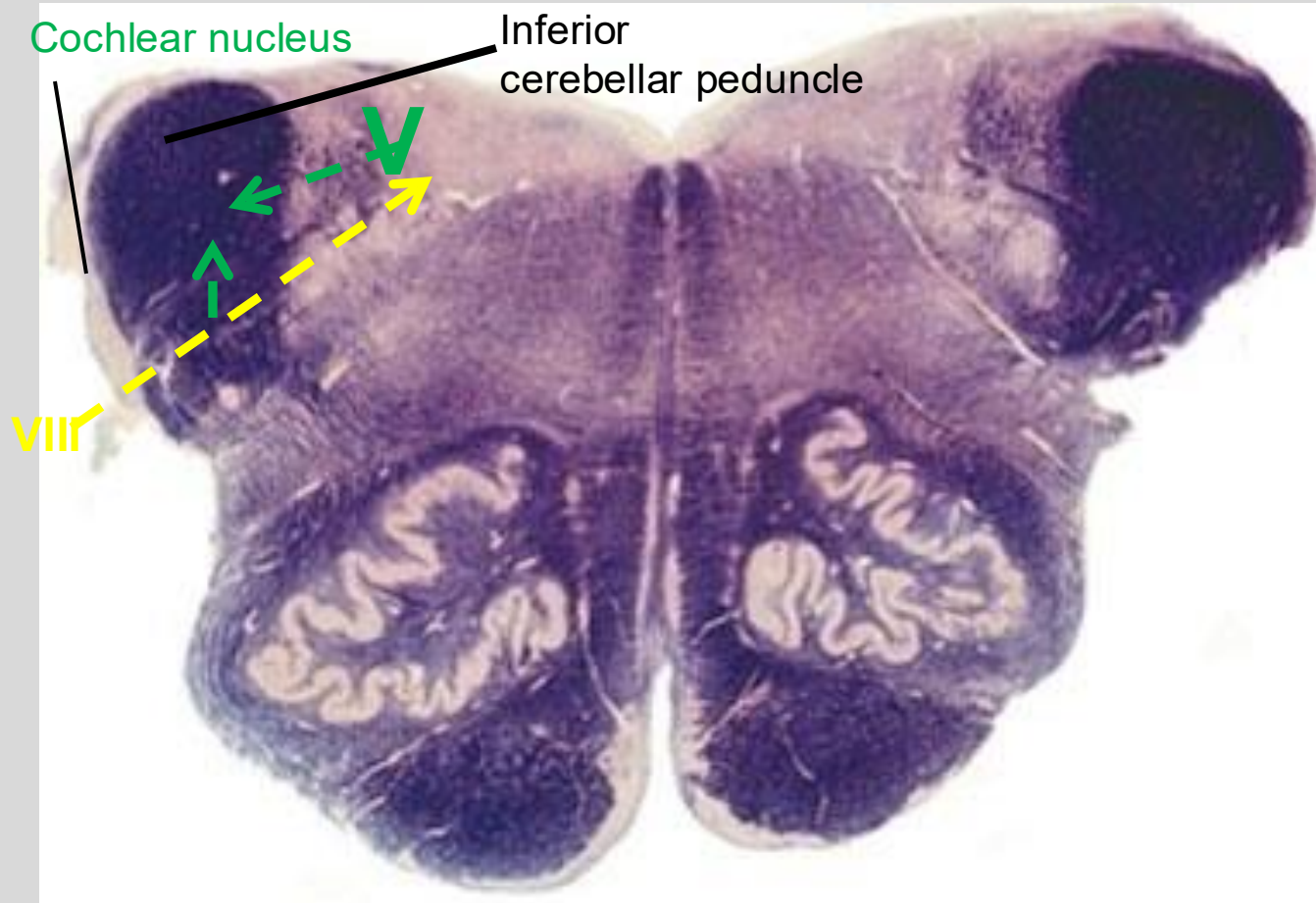
vestibular nucleus ← semicircular canals, utricles & saccules (head orientation)

Vestibular nucleus is massive and contributes projections to the cerebellum, medial longitudinal fasciculus for eye movement control, and vestibulospinal tracts for controlling head orientation, and legs.

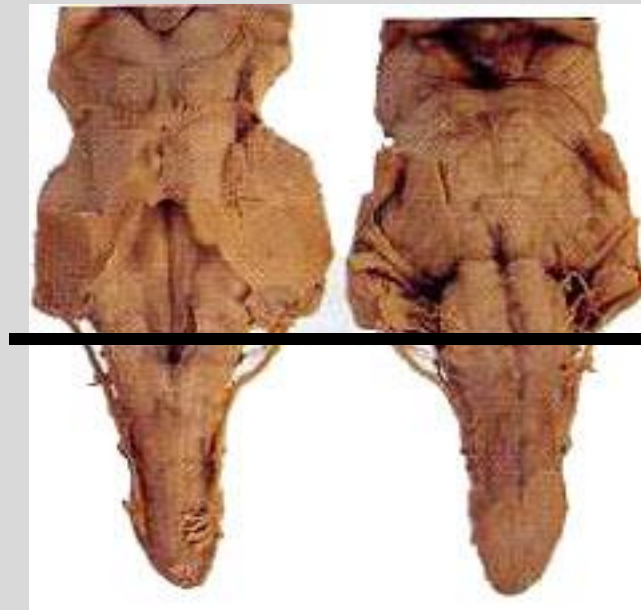
CN VIII (vestibulocochlear)

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pons/medulla
junction



VIII	vestibular	head motion / orientation
VIII	cochlear	hearing

CN VIII Clinical correlation

Excessive drinking alcohol – osmotic imbalance in semi-circular canals leads to sensation of spinning, i.e. “vertigo”.

Meniere’s disease – triad of diminished audition, aural fullness, and vertigo. Inflammation of canals.

Edinger-Westphal (III)

Oculomotor (III)

Trochlear (IV)

Trigeminal (V)

Abducens (VI)

Facial (VII)

Superior (VII)

and inferior (IX)
salivatory

Ambiguus (IX, X, XI)

Dorsal motor nucleus
of vagus (X)

Hypoglossal (XII)

Spinal accessory (XI)

Motor nuclei

Trigeminal
(V, VII, IX, X)

Sulcus limitans

Vestibular
(VIII)

Cochlear
(VIII)

Solitary (VII, IX, X)

Sensory nuclei

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CN IX (glossopharyngeal)

CN IX (glossopharyngeal)

nucleus ambiguus → *stylopharyngeus* muscle— swallowing (pharyngeal phase)

inferior salivatory nucleus → parotid gland secretion

gustatory nucleus ← posterior 1/3 of the tongue

solitary nucleus ← baroreceptor feedback from carotid sinus and body

trigeminal nucleus ← posterior 1/3 tongue, throat, and ear

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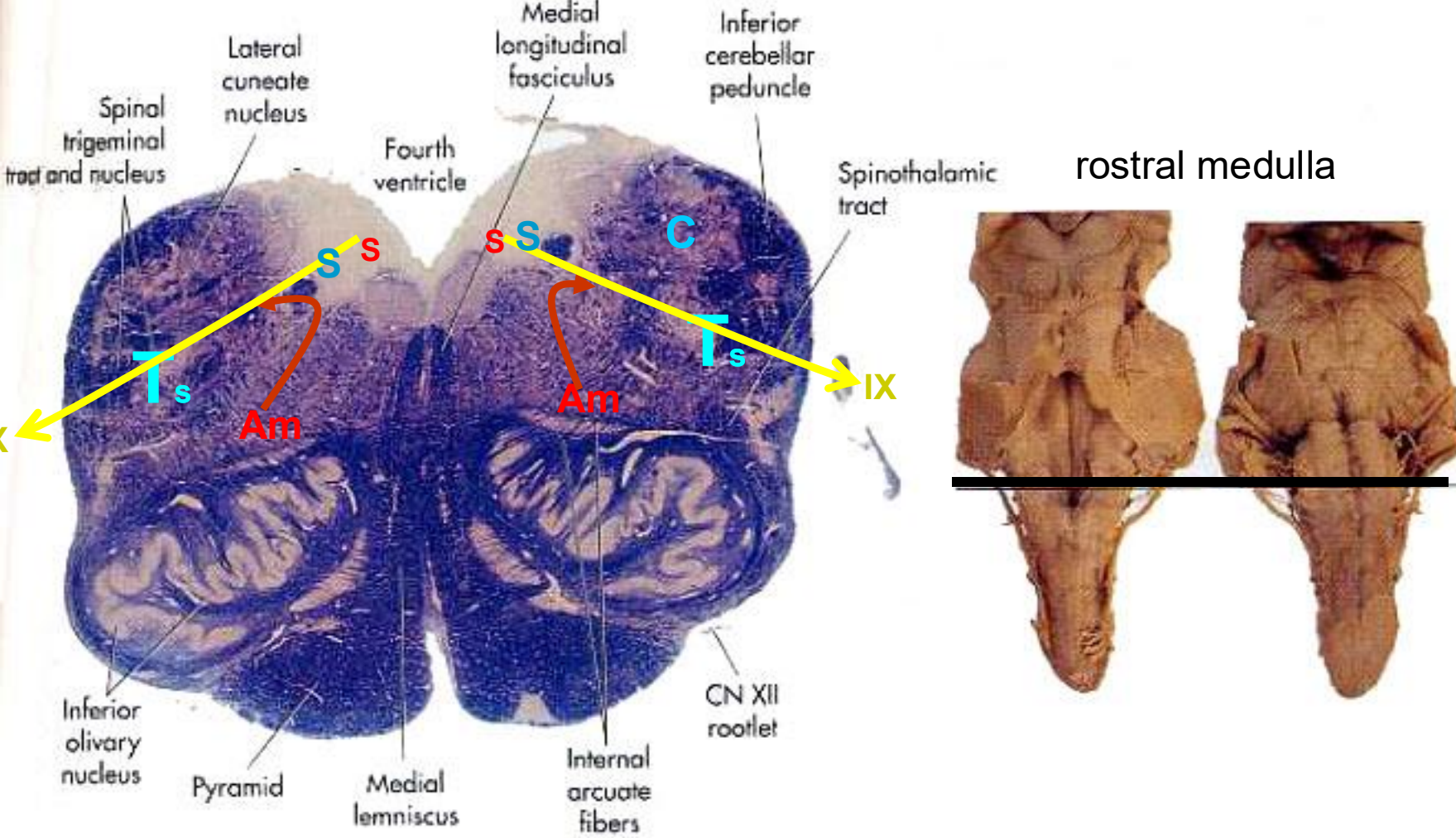
solitary nucleus ← baroreceptor feedback from carotid sinus and body

★ **trigeminal nucleus** ← posterior 1/3 tongue, throat, and ear

Remember the name: **glosso-pharyngeal**

tongue/speech e.g. glossary,
glossolalia (speaking in tongues)

pharynx



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CN IX (glossopharyngeal)

IX	ambiguus	swallowing
IX	inferior salivary	parotid gland reflex
IX	gustatory	taste (posterior 1/3)
IX	solitary	blood pressure, [O ₂] feedback
IX	trigeminal	throat, tongue, ear somatosensation

nucleus ambiguus → **stylopharyngeus muscle**

inf salivatory → **parotid gland**

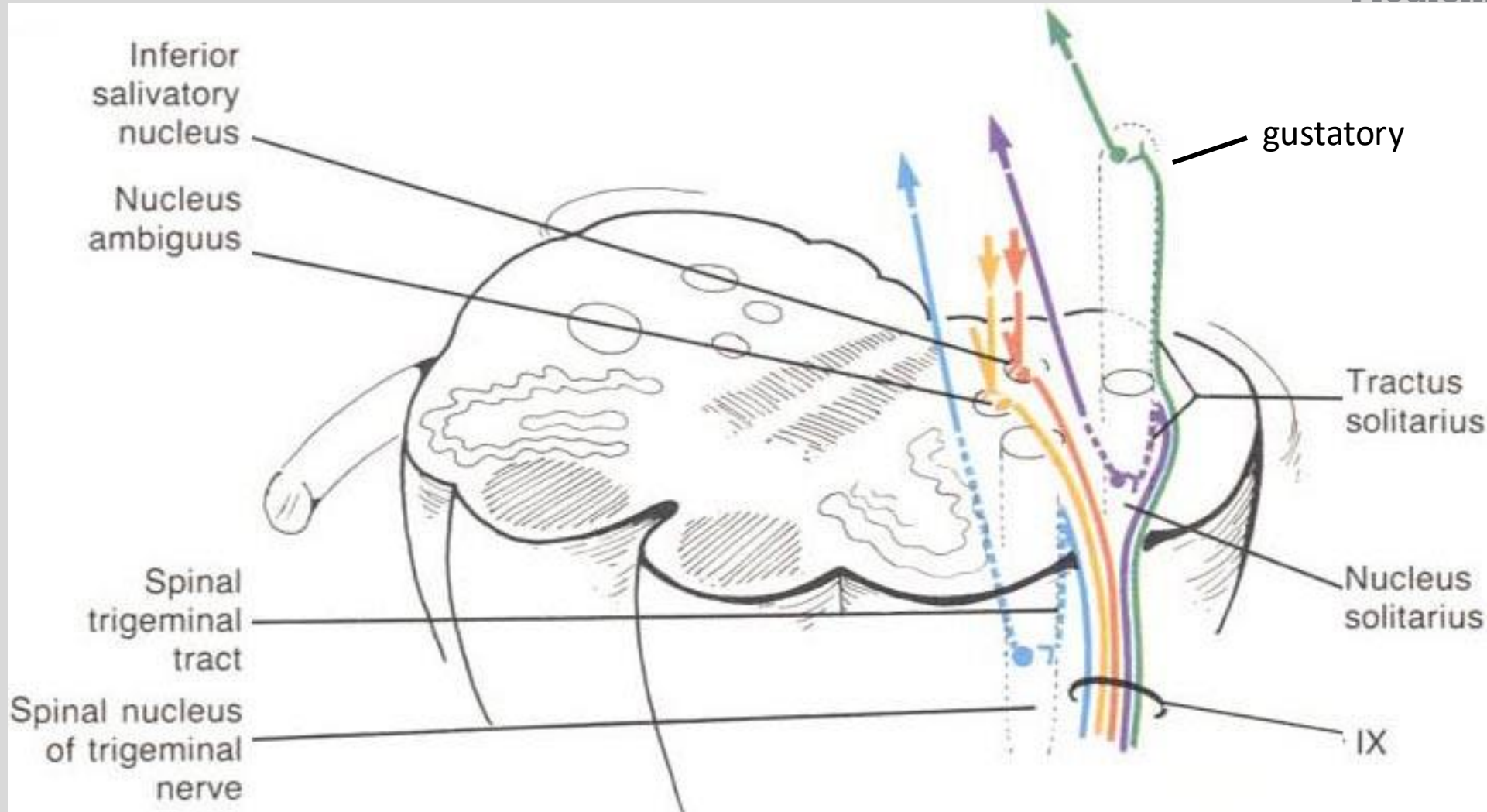
gustatory ← **posterior tongue** (note switch in color from other figs)

solitary ← **carotid sinus and body**

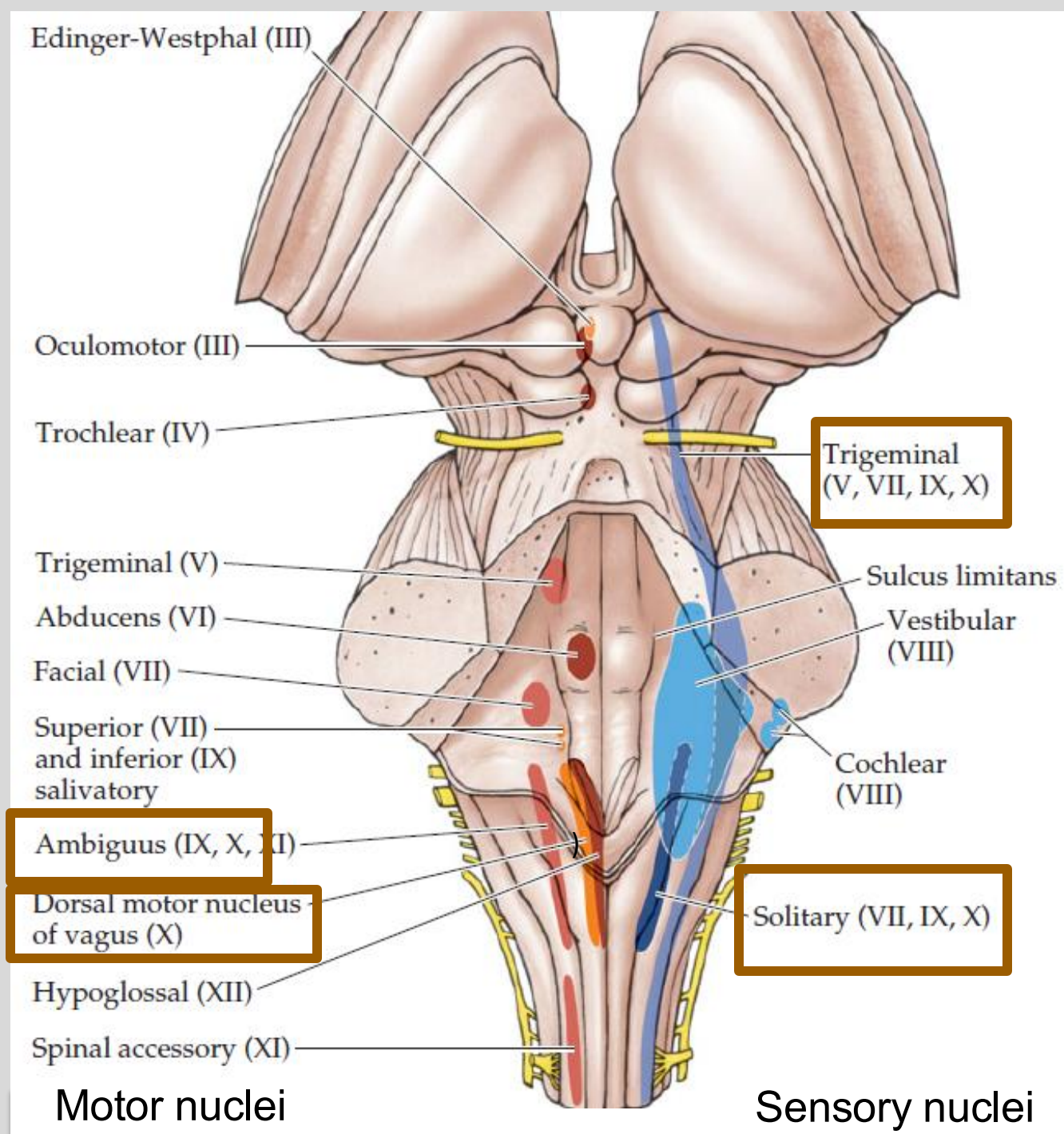
trigeminal ← **posterior tongue, throat, and ear**

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CN X (vagus)



CN X (vagus)

nucleus ambiguus → striated muscle of soft palate, pharynx (gag reflex),
and muscles for vocal cord in larynx

dorsal motor nucleus → many parasympathetic responses , “rest and digest” incl.
slower heart rate and increased secretions and motility of the GI tract

solitary nucleus ← visceral sensory from larynx, esophagus, trachea, (i.e. mucous membranes),
abdominal & thoracic viscera & aortic arch stretch receptors, aortic bodies
(chemoreceptors) (Nucleus of Solitary Tract, NST)

trigeminal nucleus ← skin of the back of ear and ear canal

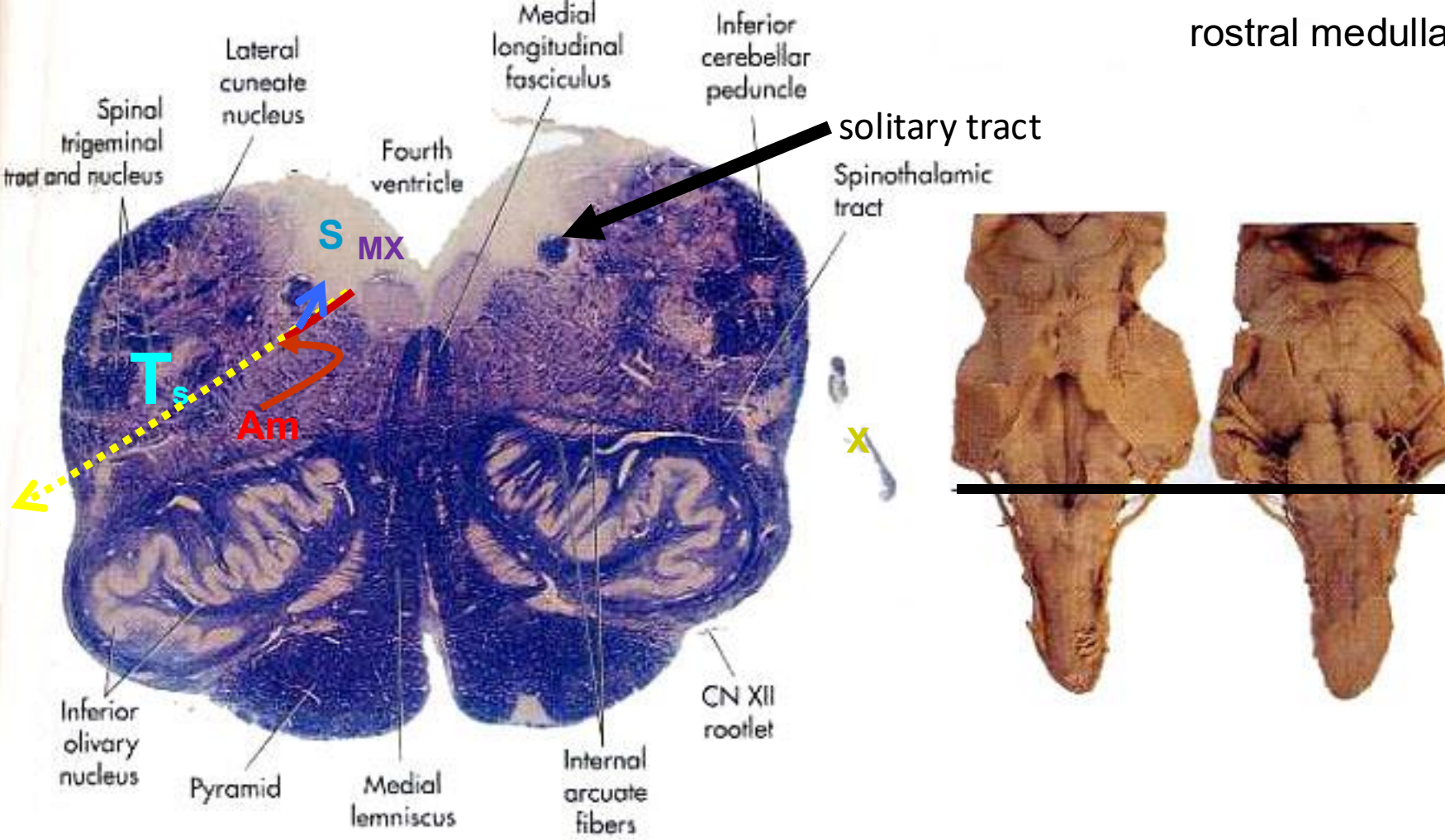
The vagus nerve is the largest conduit of autonomic output from the brainstem
AND autonomic input to the brainstem. In fact, 65-80% are autonomic afferents!

rostral medulla

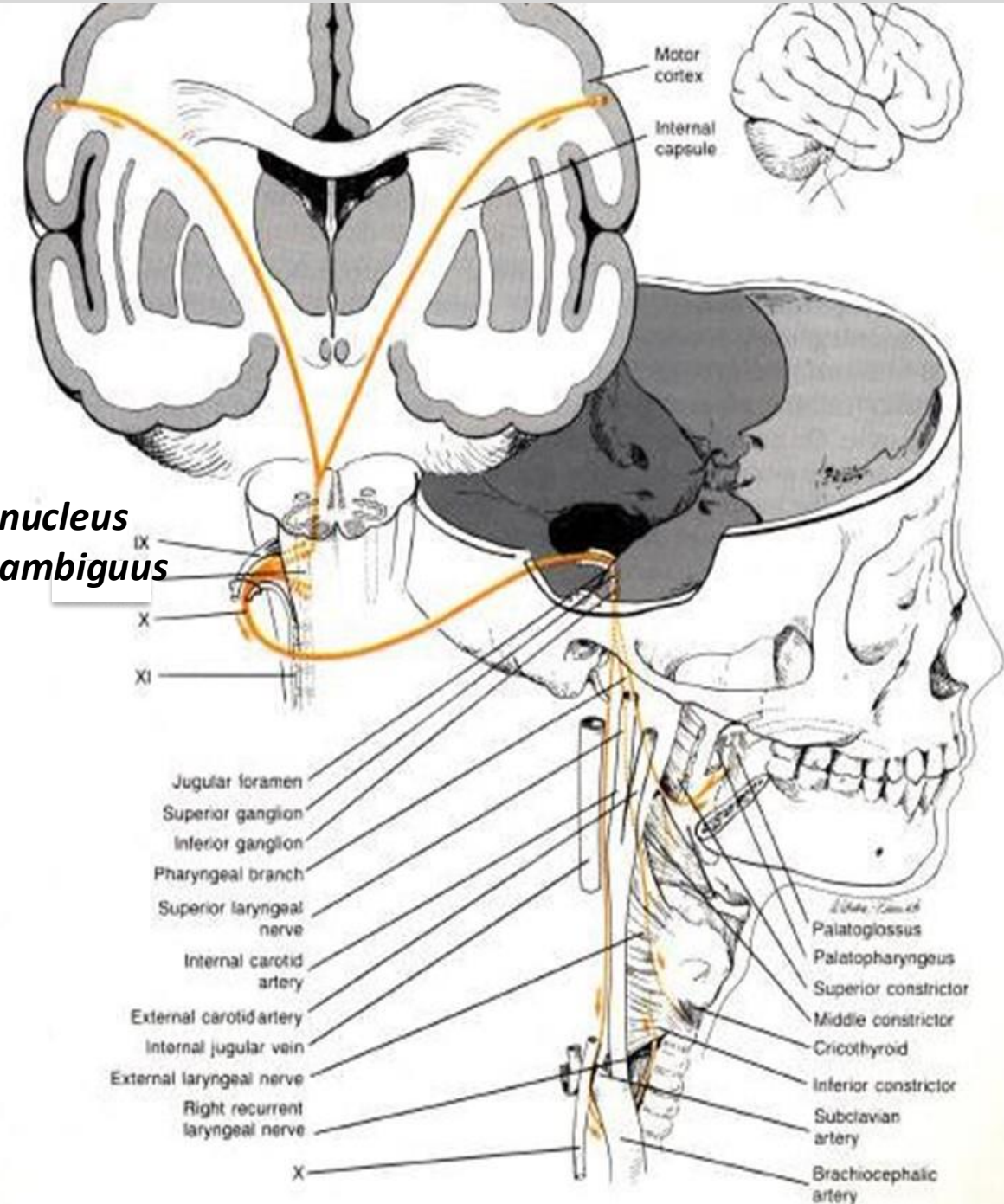
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CN X (vagus)



- | | | |
|---|---------------------------|--|
| X | ambiguus | throat, vocals motor, <i>and heart efferents</i> |
| X | dorsal motor vagus | gastrointestinal, heart efferent |
| X | solitary | respiratory, cardio, digestive feedback |
| X | trigeminal | ear somatosensation |



CN X (vagus)

nucleus ambiguus
projects to **ipsilateral**
throat muscle

unilateral medulla
lesion will cause
ipsilateral paralysis

the nucleus ambiguus
is **bilaterally** controlled
by upper / cortical
motor neurons

unilateral cortical
lesion will be largely
compensated

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CN X Clinical correlation

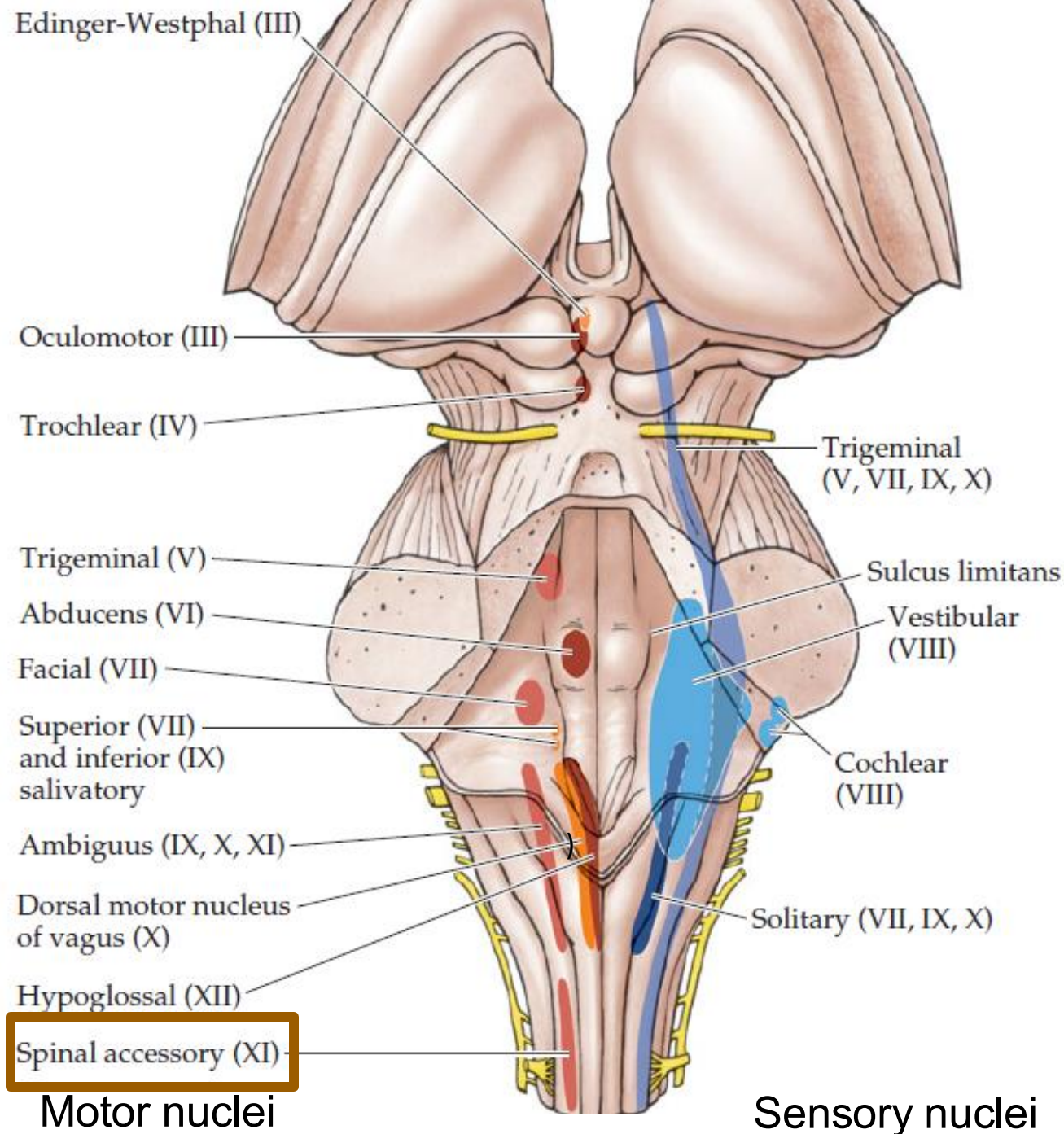
CN X-compression — Aortic arch aneurysm or tumors in neck/throat

Ipsilateral paralysis of soft palate, pharynx, and larynx

→ Hoarseness, deviated uvula, difficulty swallowing, and loss of gag (palatal) reflex

Autonomic dysfunctions include GERD and gastroparesis

CN XI (spinal accessory)



CN XI (Spinal Accessory)

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spinal accessory nuclei → *sternocleidomastoid (turn head to opposite side) and upper trapezius (raise shoulder)*

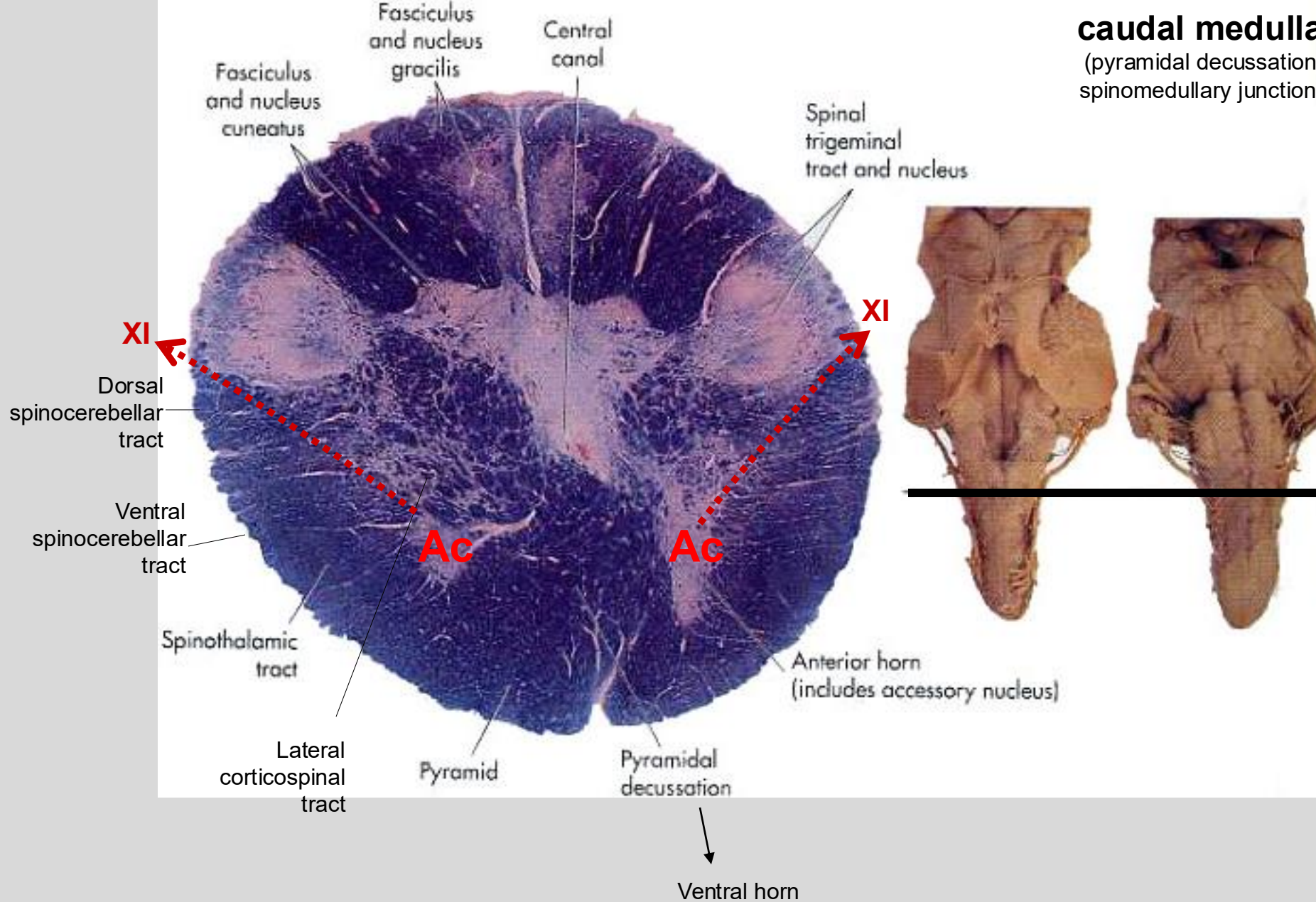
**some sources claim ipsilateral projection from cerebral cortex; we stick with bilateral.*

caudal medulla

(pyramidal decussation/
spinomedullary junction)

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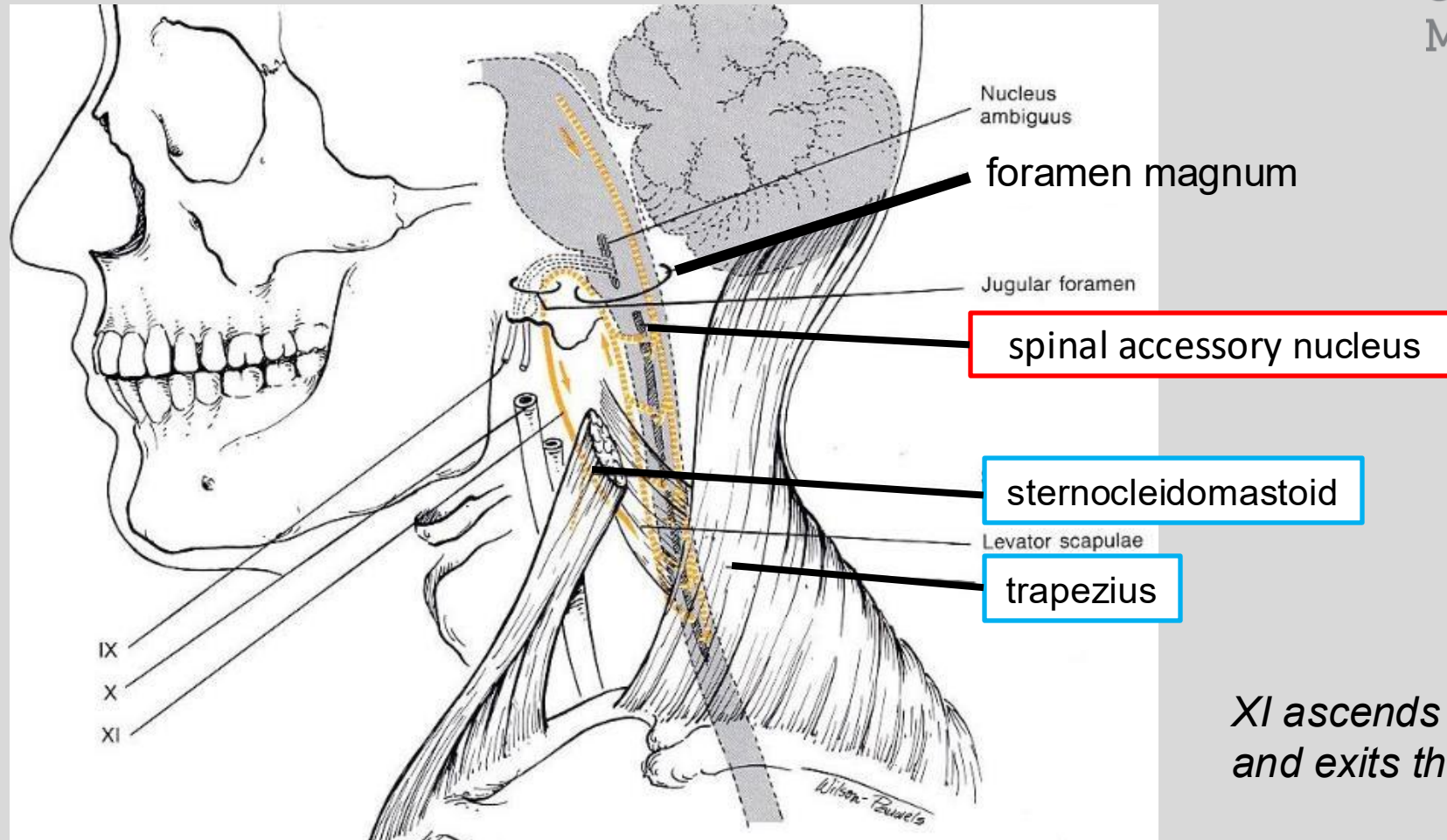


XI spinal accessory neck, shoulder

CN XI (Spinal Accessory)

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*XI ascends through foramen magnum
and exits through jugular foramen!*

CN XI Clinical correlation

Trapezius palsy – shoulder drooping, winged scapula

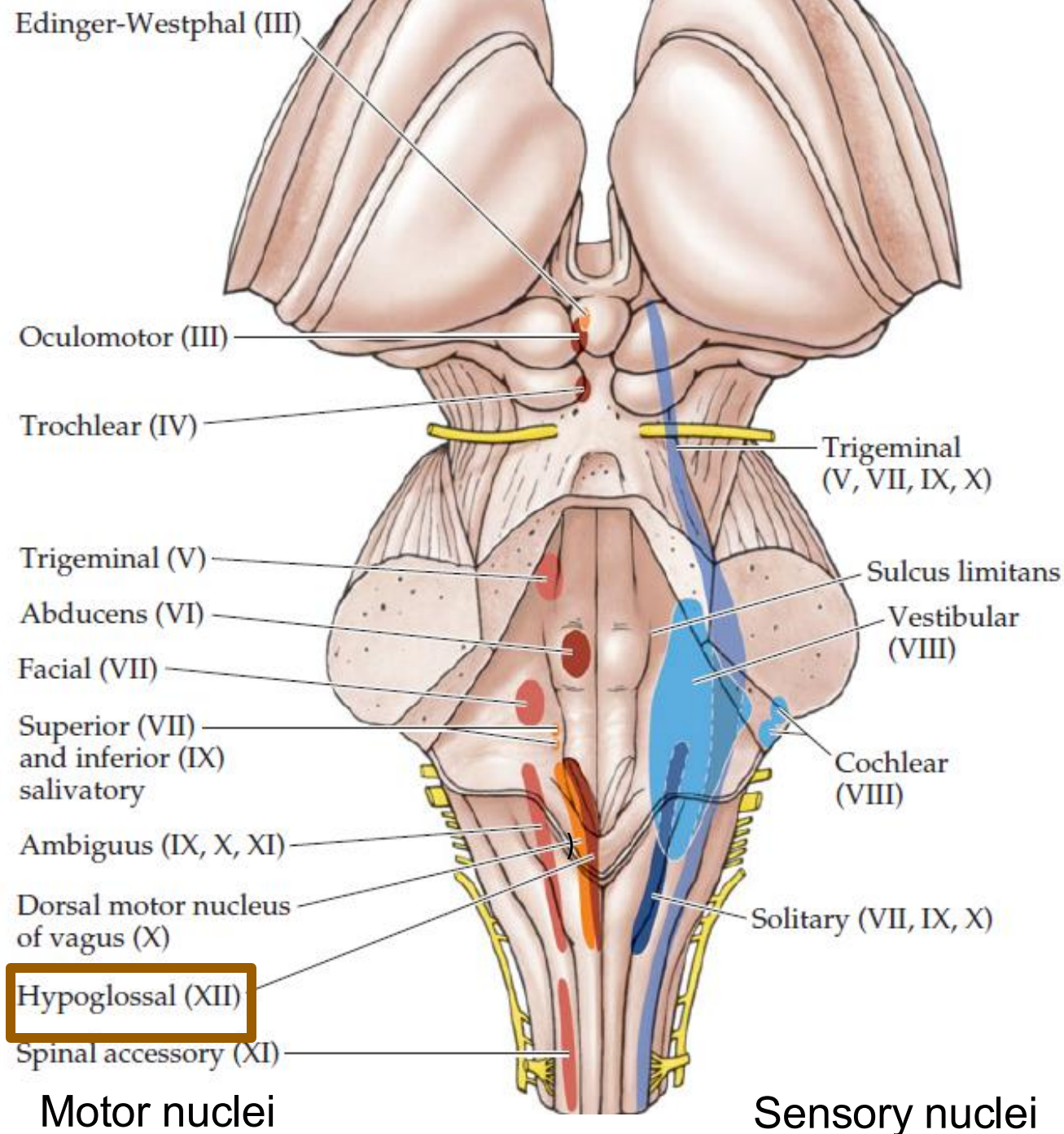
Caused by:

trauma to the posterior cervical triangle
iatrogenic injury -> caused by surgery
(e.g. lymph node biopsy)



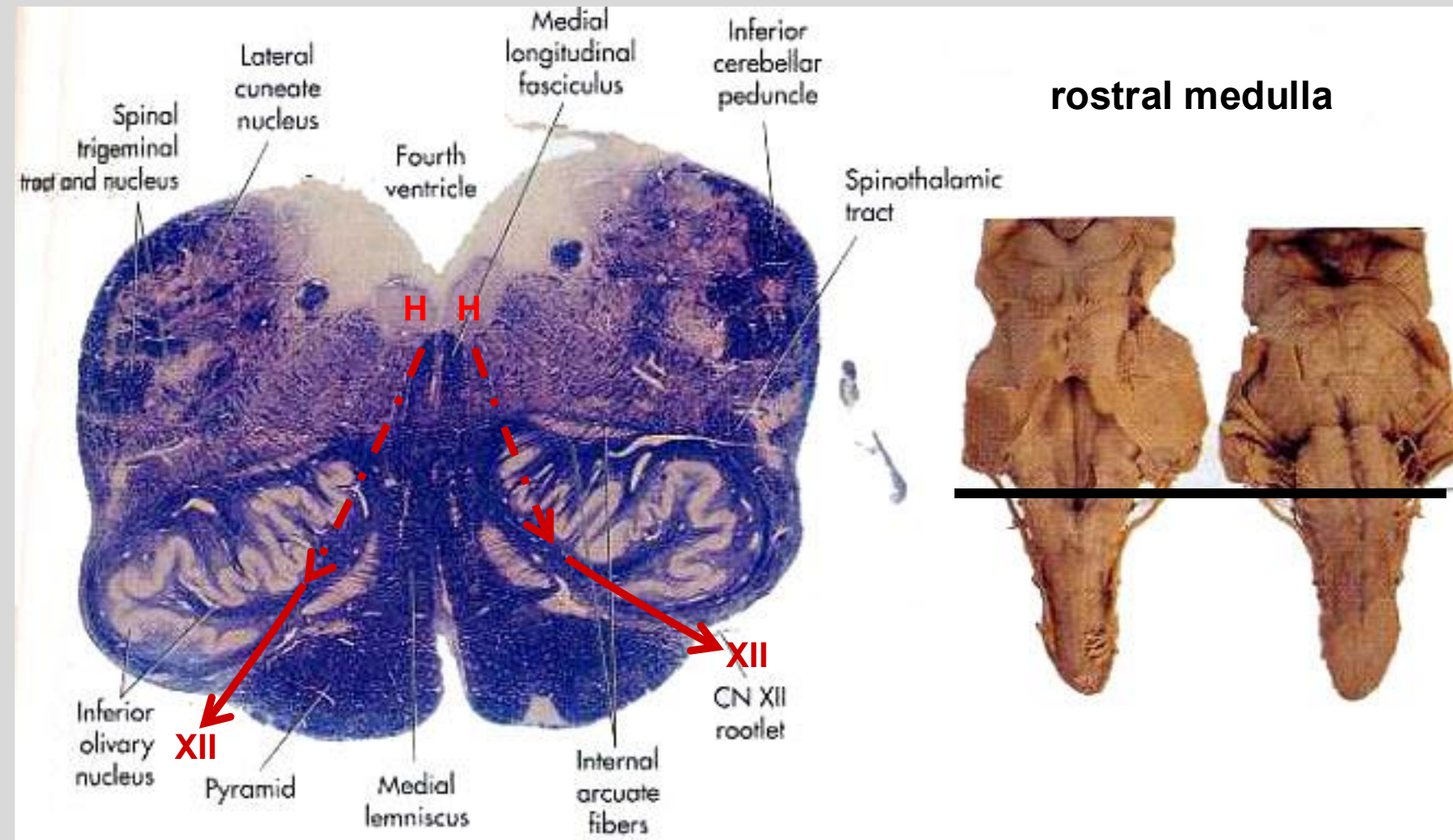
Kelley et al. (2008) Spinal accessory nerve palsy: associated signs and symptoms.

CN XII (hypoglossal)



CN XII (hypoglossal)

rostral medulla



XII hypoglossal tongue

CN XII (hypoglossal)

hypoglossal nuclei → all the internal and most of the external muscles of the tongue

CN XII (hypoglossal)

Hypoglossal nerve palsy

Destruction of the hypoglossal nucleus, or interruption of the motor axons

Tongue deviates to the weak side
Due to unopposed action of contralateral *genioglossus* muscle

Muscle atrophy

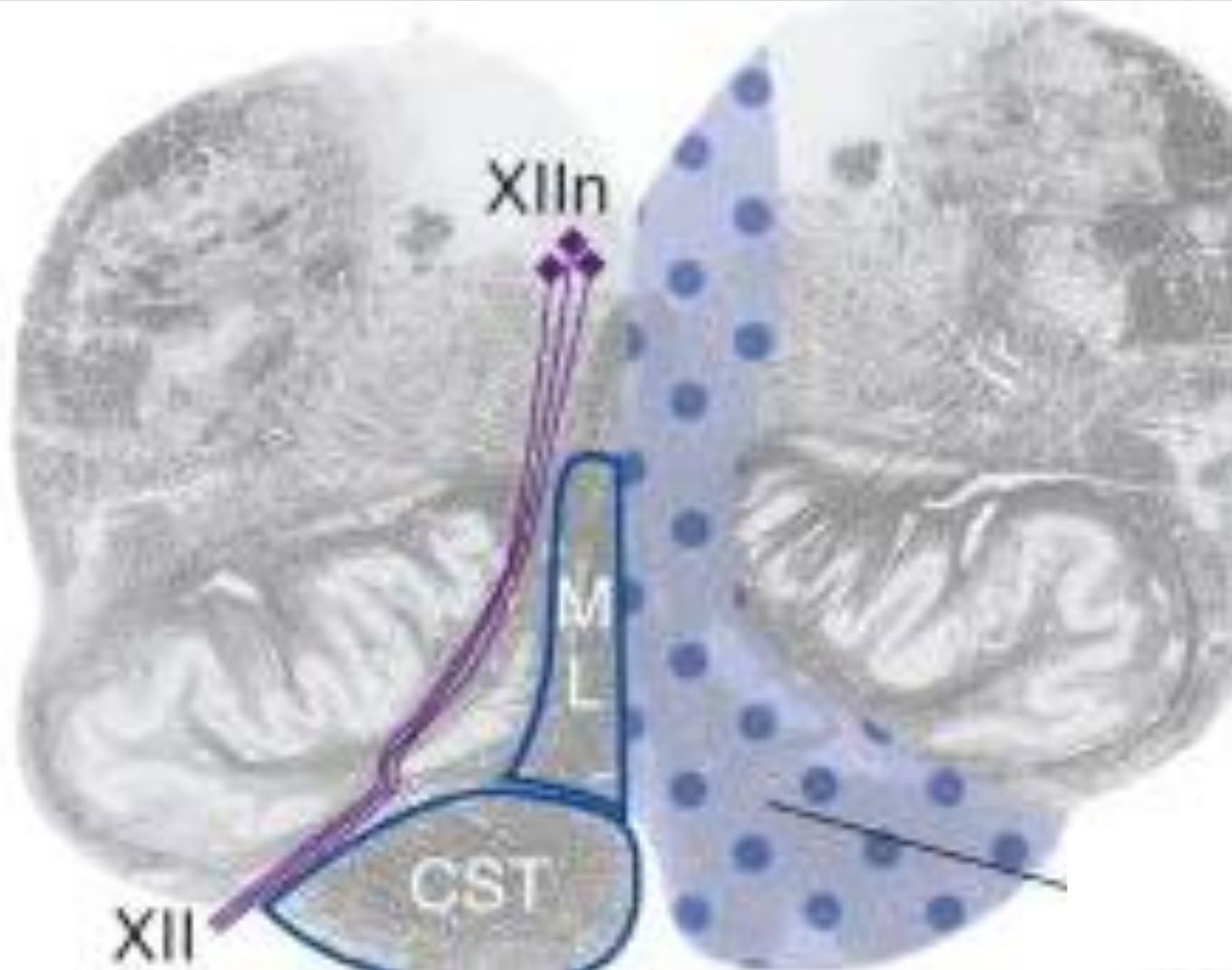
Isolated damage is relatively rare:
Tumors, iatrogenic, penetrating trauma.



Medial medullary syndrome

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Loss of vascular supply
(Anterior spinal artery)

Damage to:

Pyramid/CST - contralateral hemiparesis of
body

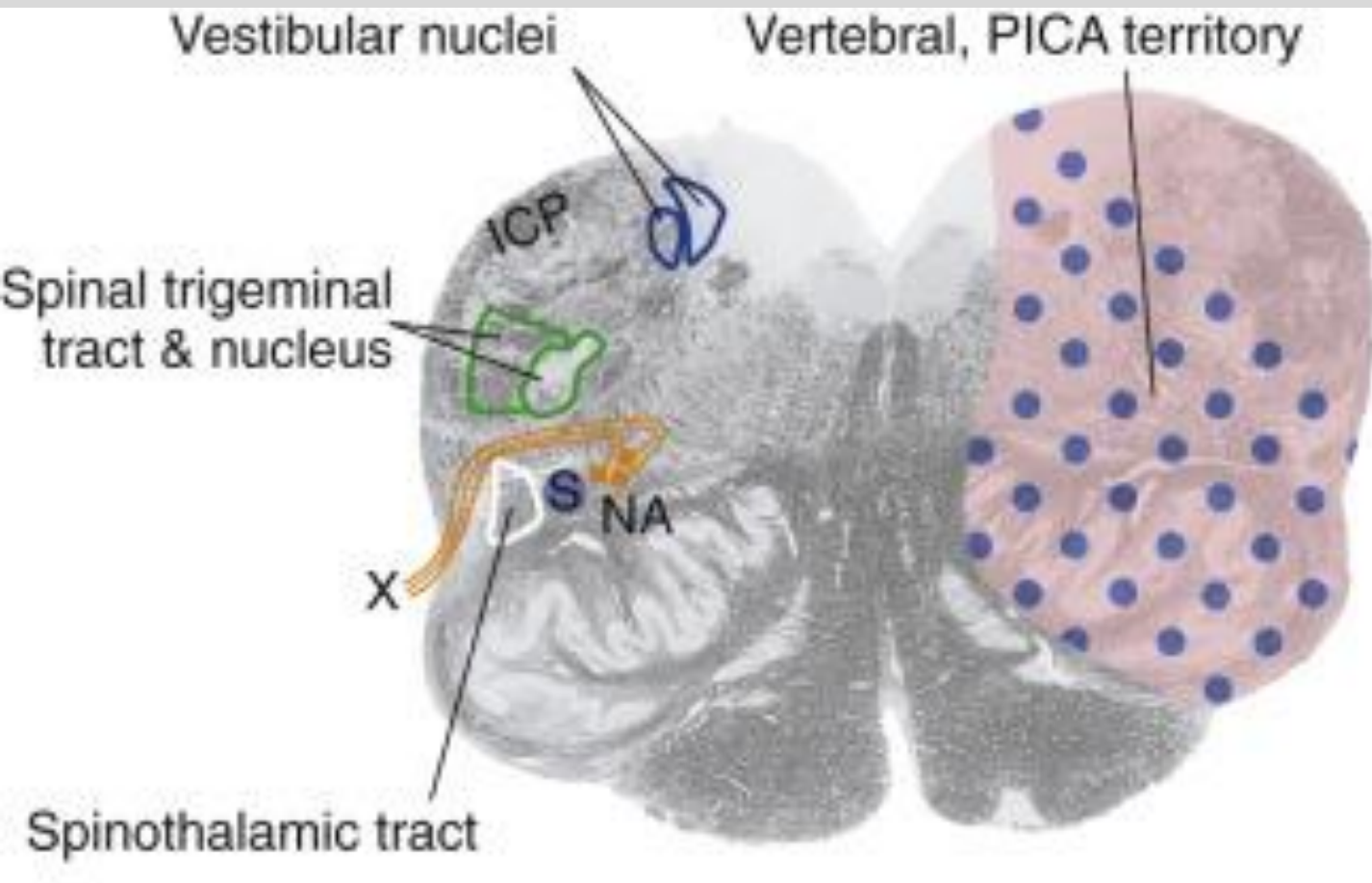
Medial lemniscus/ML - contralateral tactile
and proprioceptive deficit of body

Hypoglossal nucleus (XIIIn) – tongue points
to ipsilateral side

Lateral medullary syndrome

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Loss of vascular supply
(posterior inferior cerebellar artery)

Damage to:

Spinothalamic tract - contralateral pain and temp deficit of body

Spinothalamic nucleus (S) - ipsilateral pain and temp deficit of face

Nucleus ambiguus (NA) - hoarseness and difficulty in swallowing

Vestibular nucleus – falling to the ipsilateral side

Inferior cerebellar peduncle (ICP) –
dyscoordination on ipsilateral leg and arm

Summary

- There is a rational medial-lateral layout of the cranial nerve nuclei.
- The pontine trigeminal nucleus processes touch & proprioception. The spinal trigeminal nucleus processes pain & temperature. Most afferents are from the trigeminal nerve. Others from facial, glossopharyngeal, and vagus nerve.
- The facial nerve relays motor efferents from the facial motor nucleus and superior salivary nucleus. Relays afferents to gustatory nucleus and trigeminal nuclei.
- The vestibulocochlear nerves uniquely supply the afferents to the vestibular and cochlear nuclei.
- The glossopharyngeal nerve is aptly named. The ambiguous nucleus controls swallowing, the gustatory nucleus receives tongue taste, and tongue/throat somatosensation is sent to the trigeminal nucleus.
- The vagus relays autonomic input to the solitary nucleus and output from the dorsal motor vagus. It also relays pharyngeal and laryngeal control from the ambiguous nucleus.
- Hypoglossal and spinal accessory nerves are motor efferents uniquely from the hypoglossal and spinal accessory nuclei.

Feedback is appreciated!

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

